

V4.2 Toy Model 5C

Coupling Scale and Baryonic Tully-Fisher Check

Rev A - Dark Sector / Scaling-Law Stress Test

Date: 2026.05.26 | Continuation after Toy Model 5B

Status: Toy Model 5C tests whether the 5B annular graph result can reproduce the baryonic Tully-Fisher scaling. It finds that 5B's shape is enough for flat curves, but not enough for the mass-scaling law unless a second constitutive rule supplies a square-root coupling to baryonic flux. This does not fit real galaxy data, derive the physical acceleration scale, disprove dark matter, or prove physical 3T.

1. Purpose

Toy Model 5B derived the flat rotation-curve shape from radial annular graph edge economics. The remaining question is amplitude scaling: as baryonic mass changes from galaxy to galaxy, does the flat velocity follow the baryonic Tully-Fisher relation?

The baryonic Tully-Fisher target is:

$$v_{\text{flat}}^4 \propto M_b$$

2. Starting point inherited from 5B

5B gave this result outside the visible source in an effective 2D annular graph:

$$r \, d\Phi/dr = \text{constant}$$

$$|d\Phi/dr| \propto 1/r$$

therefore:

$$a_{\text{rel}} \propto 1/r$$

$$v_{\text{rel}}^2 = r \, a_{\text{rel}} = \text{constant}$$

This derives the flat curve shape. But the amplitude of that constant still has to be linked to the baryonic mass M_b .

3. First test: constant coupling fails BTF

If the same coupling Γ applies to every galaxy, then the 5B outer velocity gives:

$$v_{\text{flat}}^2 = \Gamma \, M_b$$

therefore:

$$v_{\text{flat}}^4 = \Gamma^2 \, M_b^2$$

That gives a log-log slope of 2, not the observed BTF target slope of 1. Therefore, the 5B annular graph law alone is not sufficient to recover BTF.

4. Required coupling law

For the BTF exponent to be 1, the coupling cannot be constant. In a general power-law family:

$$\Gamma(M_b) \propto M_b^q$$

$$v_{\text{flat}}^2 = \Gamma(M_b) \, M_b$$

$$v_{\text{flat}}^4 \propto M_b^{2(1+q)}$$

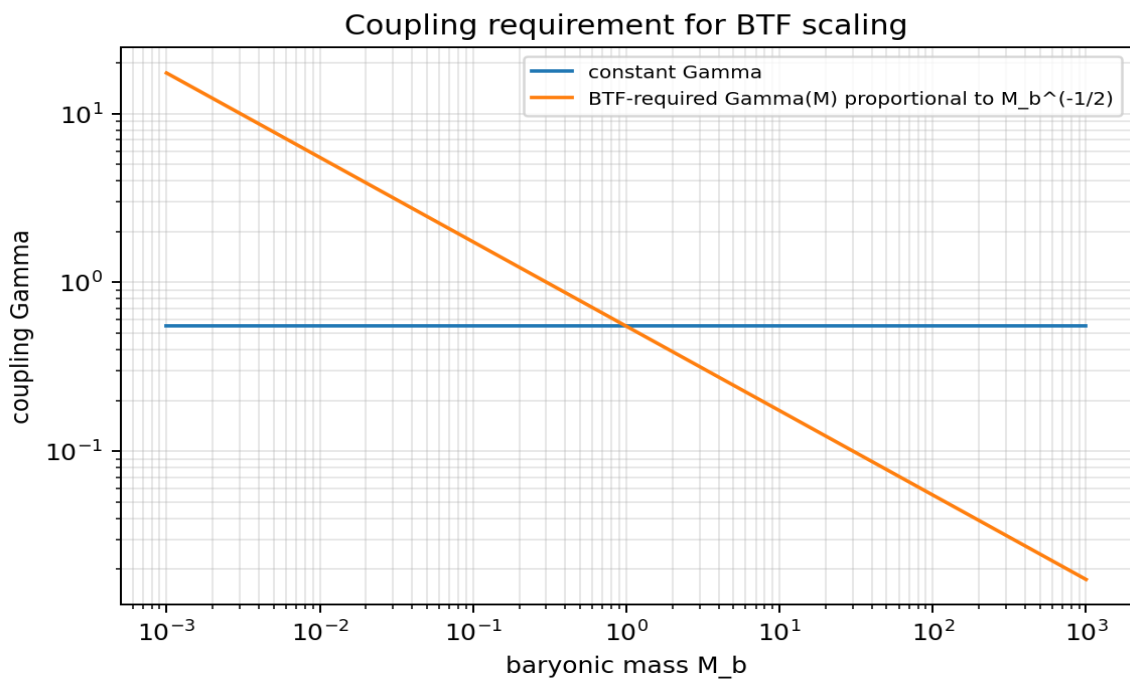
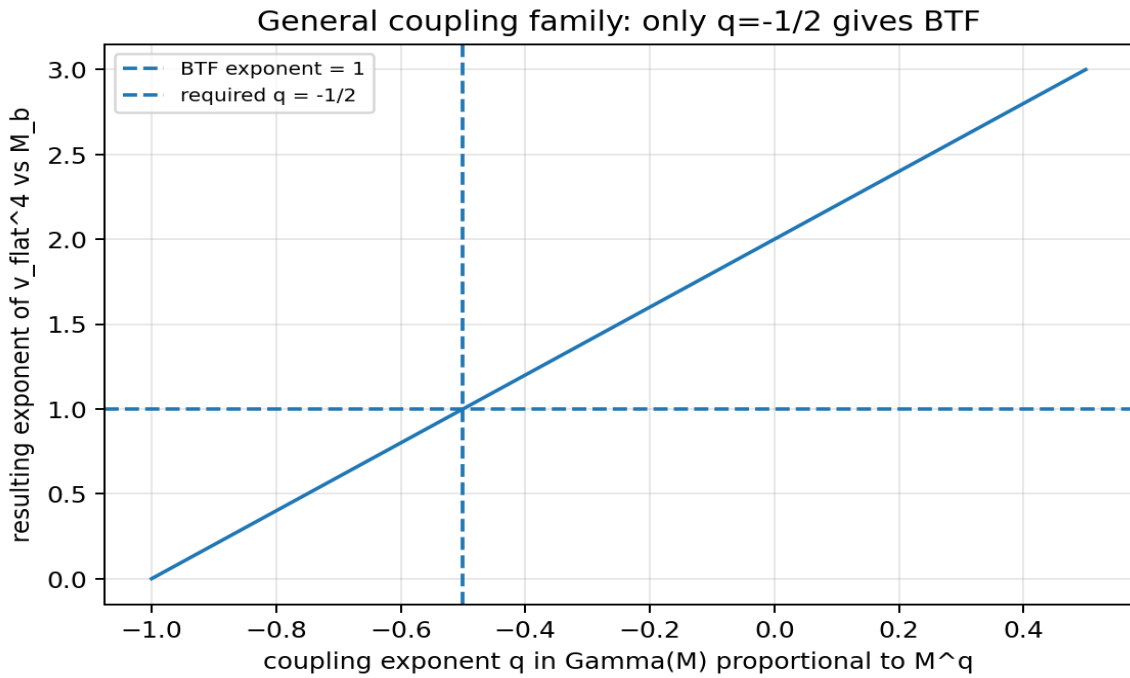
Set the exponent equal to 1:

$$2(1+q) = 1$$

$$q = -1/2$$

Therefore BTF requires:

$$\Gamma(M_b) \propto M_b^{(-1/2)}$$



5. Threshold / square-root edge-stress law

The cleanest V4.2-compatible way to write the required scaling is not as arbitrary tuning of Γ , but as a square-root edge-stress rule:

$$T_{\text{edge}}^2 = A_R G M_b$$

$$v_{\text{flat}}^2 = T_{\text{edge}} = \sqrt{A_R G M_b}$$

therefore:

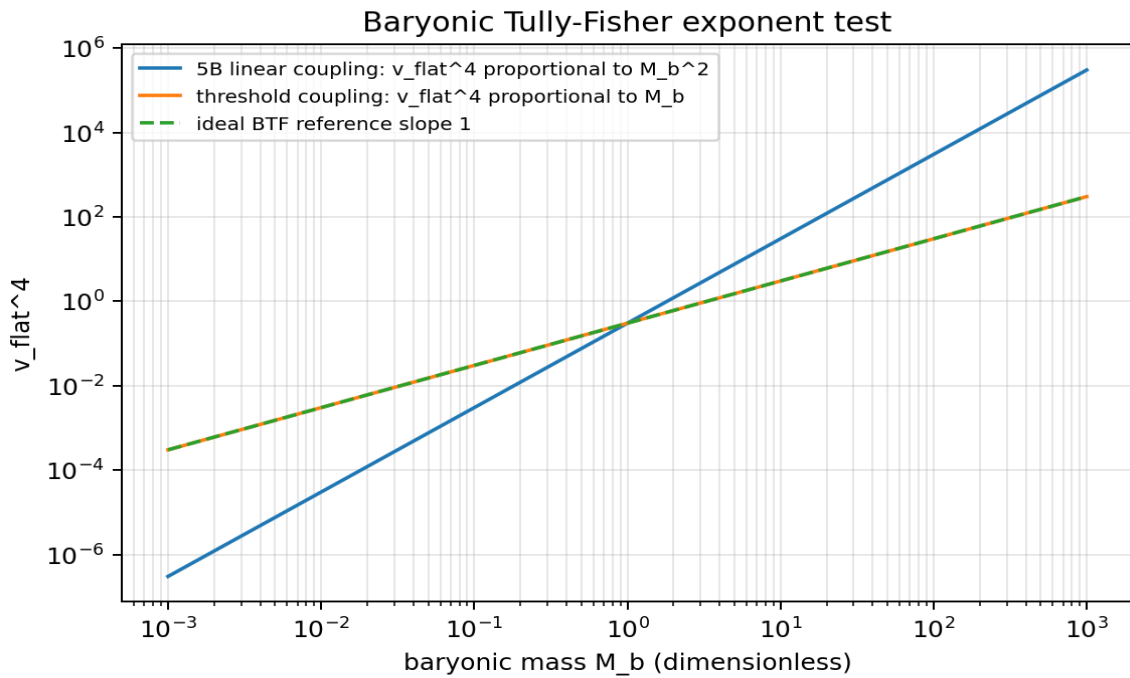
$$v_{\text{flat}}^4 = A_R G M_b$$

Meaning of symbols:

- T_{edge} is the saturated outer relational edge tension.
- A_R is a universal relational threshold scale.
- G is the gravitational coupling in the dimensionless toy model.
- M_b is the total baryonic source mass.
- This produces exactly the BTF exponent in the toy model.

Important guardrail: A_R is still an added universal scale in 5C. The model shows what scaling law is required and how it can be expressed relationally. It does not derive A_R from deeper 3T physics.

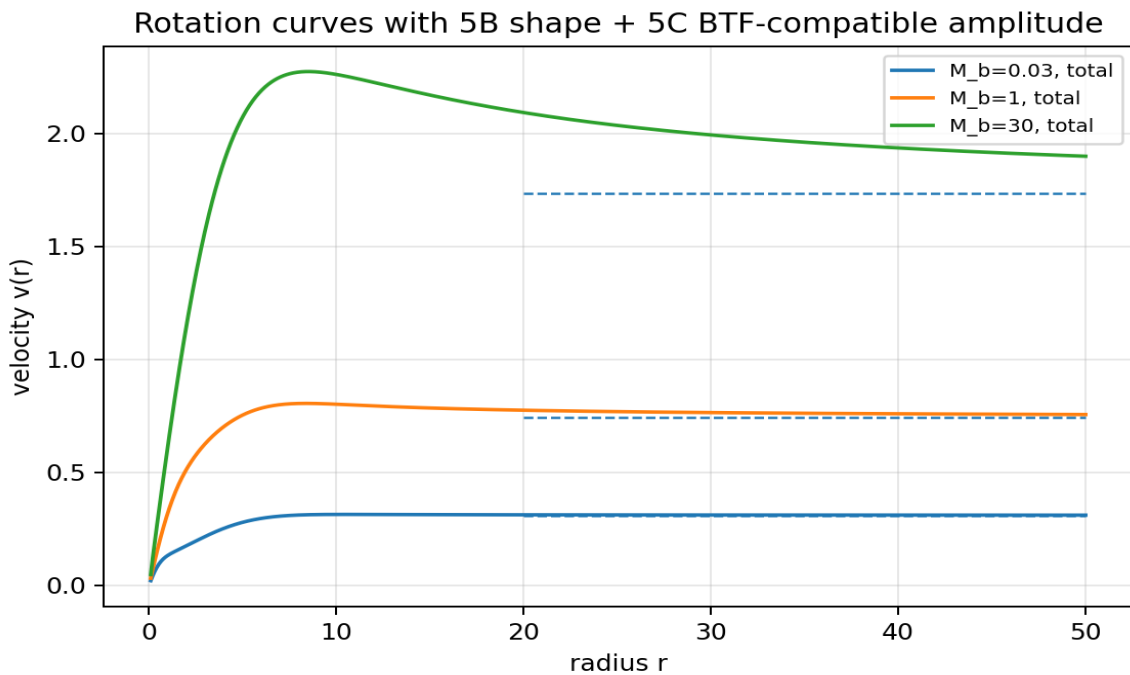
6. Scaling-law numerical check

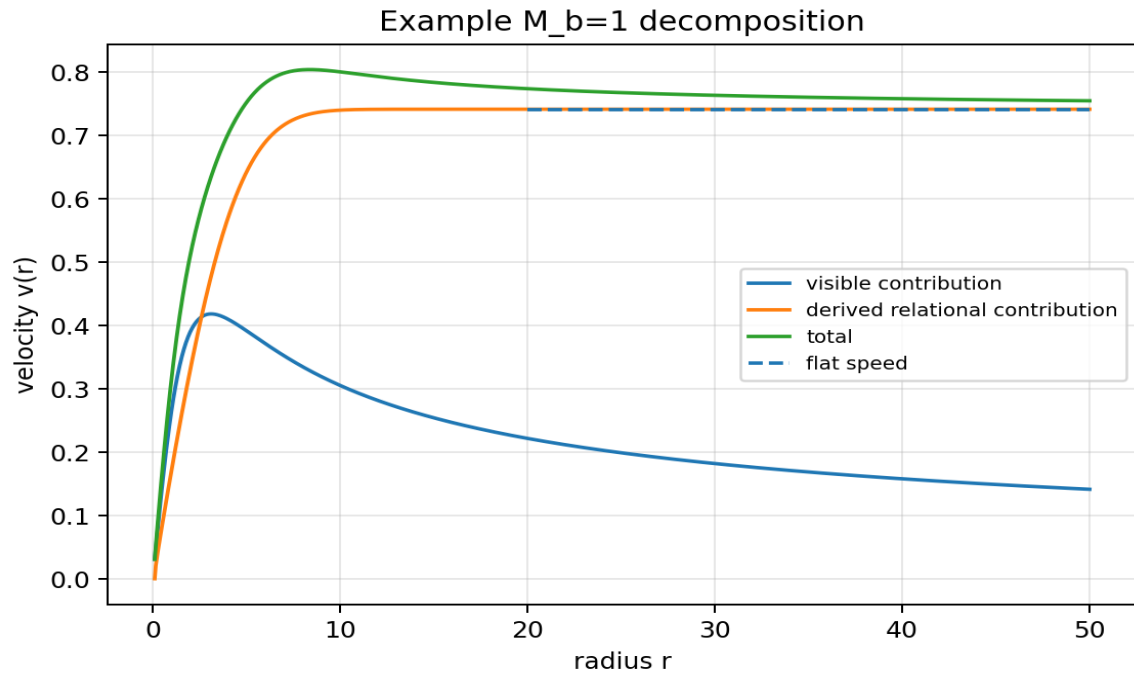


Diagnostic	Value	Interpretation
Linear-coupling BTF slope	2.000000	Fails: gives v_{flat}^4 proportional to M_b^2 .
Square-root threshold BTF slope	1.000000	Passes: gives v_{flat}^4 proportional to M_b .
Required Gamma slope	-0.500000	Gamma must scale as $M_b^{-1/2}$.
Linear log residual vs BTF	1.736386	Large systematic scaling error across mass range.
Threshold log residual vs BTF	6.970243e-17	Zero to numerical precision because law matches target.

7. Example rotation curves

The 5B radial shape is retained, while the 5C threshold law sets the flat-speed amplitude for different baryonic masses.





M_b	visible scale	v_{flat}	outer total slope	outer mean v
0.03	0.916	0.3086	-0.0051	0.3102
1	2.200	0.7416	-0.0277	0.7630
30	5.149	1.7356	-0.1083	1.9844

8. Mass sweep table

M_b	v_4 linear	v_4 threshold	Gamma required
0.001	3.0250e-07	3.0250e-04	1.7393e+01
0.01	3.0250e-05	3.0250e-03	5.5000e+00
0.1	3.0250e-03	3.0250e-02	1.7393e+00
1	3.0250e-01	3.0250e-01	5.5000e-01
10	3.0250e+01	3.0250e+00	1.7393e-01
100	3.0250e+03	3.0250e+01	5.5000e-02
1e+03	3.0250e+05	3.0250e+02	1.7393e-02

9. Interpretation inside V4.2

Toy Model 5C supports this limited ladder:

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5B annular graph action
-> flat rotation shape

plus

square-root edge-stress threshold:
 $T_{\text{edge}}^2 = A_R G M_b$ 

therefore:

 $v_{\text{flat}}^4 = A_R G M_b$ 
-> BTF-like scaling

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It does not support this stronger ladder:

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V4.2
-> physical value of  $A_R$  derived
-> all real galaxies fit
-> MOND replaced
-> particle dark matter disproved
-> physical 3T proven

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10. What Toy Model 5C achieves

- It proves that constant coupling from 5B fails the BTF exponent.
- It identifies the exact coupling scaling required: Gamma proportional to $M_b^{(-1/2)}$.

- It rewrites that requirement as a square-root relational edge-stress law.
- It shows that this threshold law produces v_{flat}^4 proportional to M_b .
- It cleanly separates the flat-curve shape problem from the amplitude-scaling problem.

11. What Toy Model 5C does not achieve

- It does not derive the universal relational threshold A_R .
- It does not derive the numerical value of the MOND-like acceleration scale.
- It does not fit real galaxy data.
- It does not prove that galaxy-scale relational networks obey the square-root stress law.
- It does not address lensing, clusters, Bullet Cluster, CMB, or structure formation.
- It does not disprove particle dark matter.
- It does not prove physical 3T.

12. Close-out statement for 5C

Toy Model 5C is a diagnostic success with an important guardrail. It shows that the 5B annular graph economics naturally provides the flat rotation-curve shape, but not the baryonic Tully-Fisher mass exponent by itself. A constant coupling gives the wrong scaling. To recover v_{flat}^4 proportional to M_b , the dark-sector relational response must include a square-root edge-stress threshold, $T_{\text{edge}}^2 = A_R G M_b$. The next task is to derive or justify A_R and the square-root law from deeper graph/3T principles rather than assuming them.

13. Recommended next chapter

Recommended next: Toy Model 5D - Origin of the Universal Relational Threshold Scale A_R .

- Do not proceed as if BTF has been fully derived.
- Try to derive A_R from cosmic boundary conditions, network saturation, or coupling $C(a)$.
- Check whether A_R can be linked to the cosmological scale without importing MOND by hand.
- Only after that should real galaxy rotation-curve fitting be attempted.

14. Rebind prompt for continuation

We are resuming after Toy Model 5C.

5B:
Derived flat rotation curve shape from effective 2D annular graph action.

5C:
Tested baryonic Tully-Fisher scaling.

Starting point:
 $v_{\text{flat}}^2 = \Gamma M_b$

If Γ is constant:
 $v_{\text{flat}}^4 = \Gamma^2 M_b^2$
This fails BTF because the exponent is 2 instead of 1.

General coupling:
 $\Gamma(M_b)$ proportional to M_b^q
 v_{flat}^4 proportional to $M_b^{2(1+q)}$

BTF requires:
 $2(1+q)=1$
 $q=-1/2$

Therefore:
 $\Gamma(M_b)$ proportional to $M_b^{(-1/2)}$

Equivalent square-root edge-stress law:
 $T_{\text{edge}}^2 = A_R G M_b$
 $v_{\text{flat}}^2 = T_{\text{edge}} = \sqrt{A_R G M_b}$
 $v_{\text{flat}}^4 = A_R G M_b$

Numerical result:
Linear-coupling slope = 2.
Square-root threshold slope = 1.
Required Γ slope = -0.5.

Critical guardrail:
5C does not derive A_R .

It does not prove the square-root edge-stress law.
It does not fit real galaxies or solve lensing/CMB/clusters/structure formation.
It does not disprove particle dark matter or prove physical 3T.

Next recommended:

Toy Model 5D - Origin of Universal Relational Threshold Scale A_R.